BEE HEALTH EARLY WARNING SYSTEM

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Honeybees are essential to the agricultural economy.

Recent survey shows that about 62% of commercial colonies collapsed last year due to factors like climate change, pesticide exposure etc.

Goal is to create a model that predicts hive health using information about past health parameters and weather conditions.

This model gives commercial beekeepers, tech companies, and hobbyists actionable diagnostics to proactively protect bee populations.

Feature Extraction

Data Sources

Bee Hive Health (Lower et al., 2024)* for apiaries in UT and NC collected between 2016 - 19.

Daily weather Summaries, NOAA for weather stations in UT and NC.

Raw Dataset Features

Inspection Date,
Brood Status, Bee Status,
Queen Status, Food Status,
Stressors Status,
Space Status,
Health Status

Average wind speed (mph),
Precipitation (in),
Snowfall (in),
Max/Min/Avg temperature (F)

Engineered Features

Lag Health Features
Previous inspection Brood, Queen, Bee, Food, Stressors and Space Status

Temporal Features
Days since last inspection
Hive age
Is first day of inspection

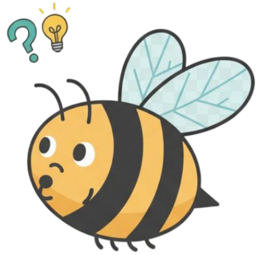
7-Day trailing average
Precipitation, Wind speed, max/min/average temperature,
Snowfall,
Number of frost days.

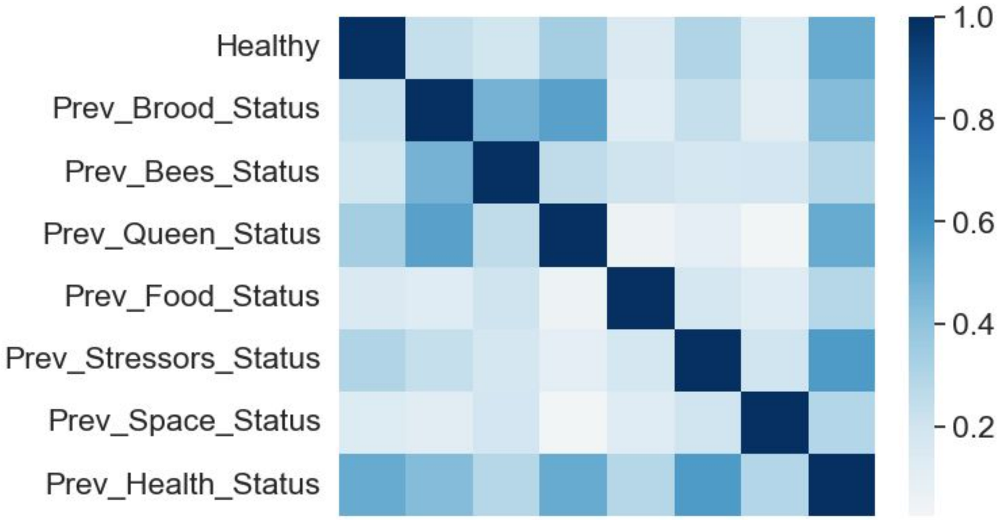
Final Features

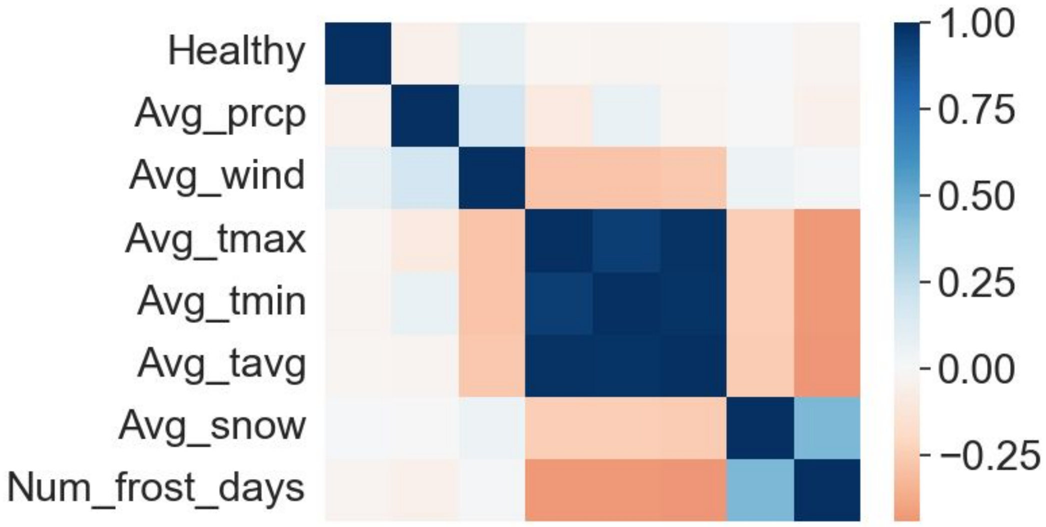
Inspection Date
Lag Health Features
Temporal Features
7- Day Trailing Average
Health Status

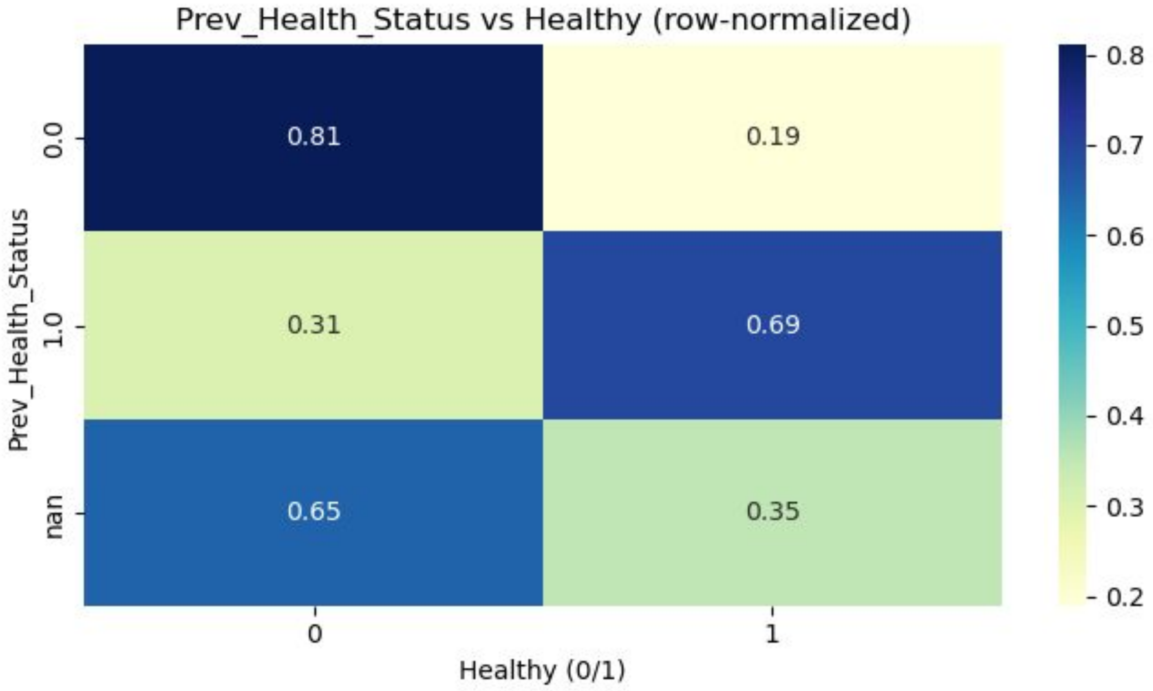
* Lower, E., Kollaparthi, S. P., Rogers, R., Hassler, E., & Cazier, J. (2024). Predicting Honeybee Health: The Healthy Colony Checklist,Hive Scale and Weather Data. *Data & Analytics for Good*. https://data-for-good.pubpub.org/pub/rg3364dl

Exploratory Data Analysis

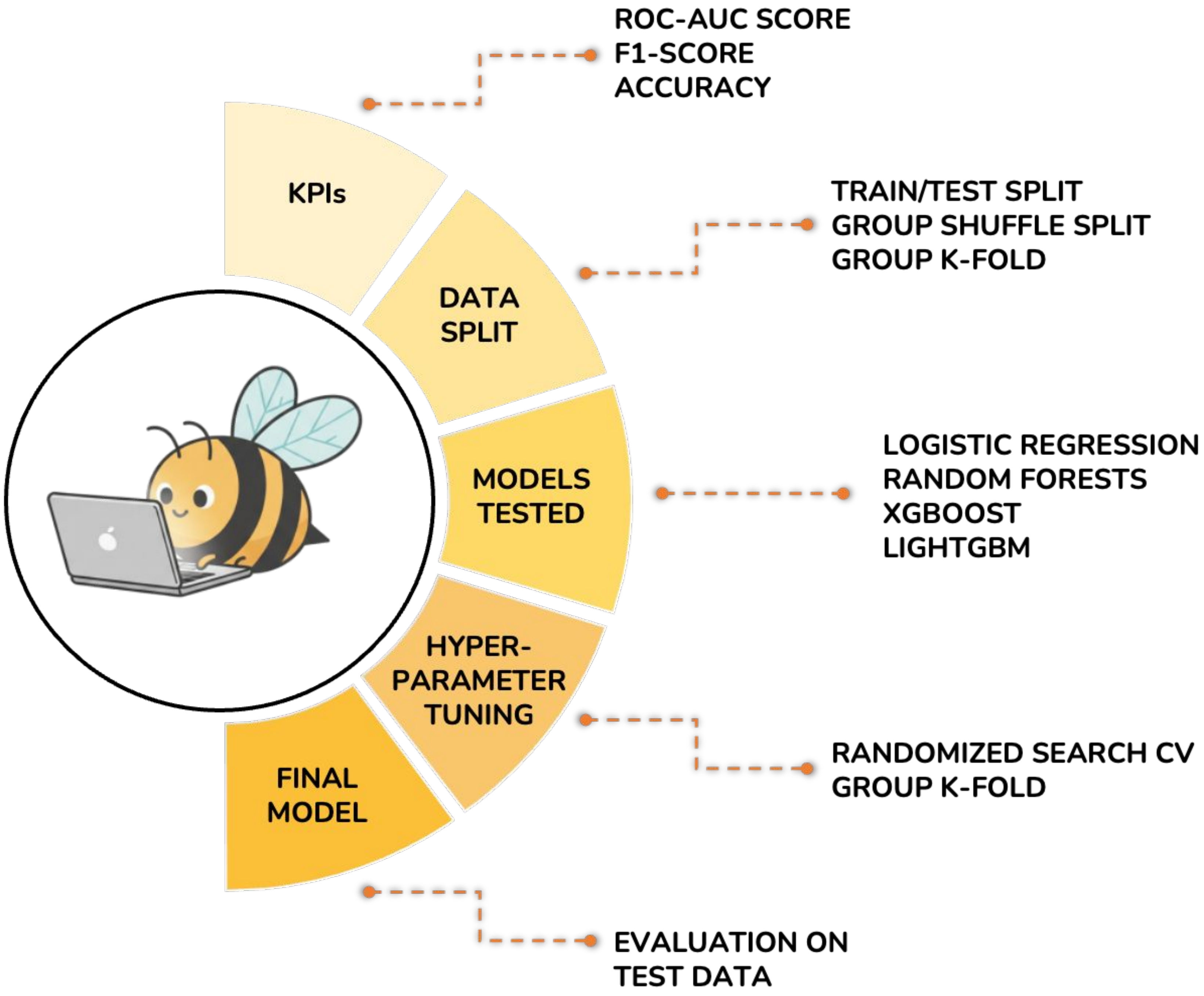






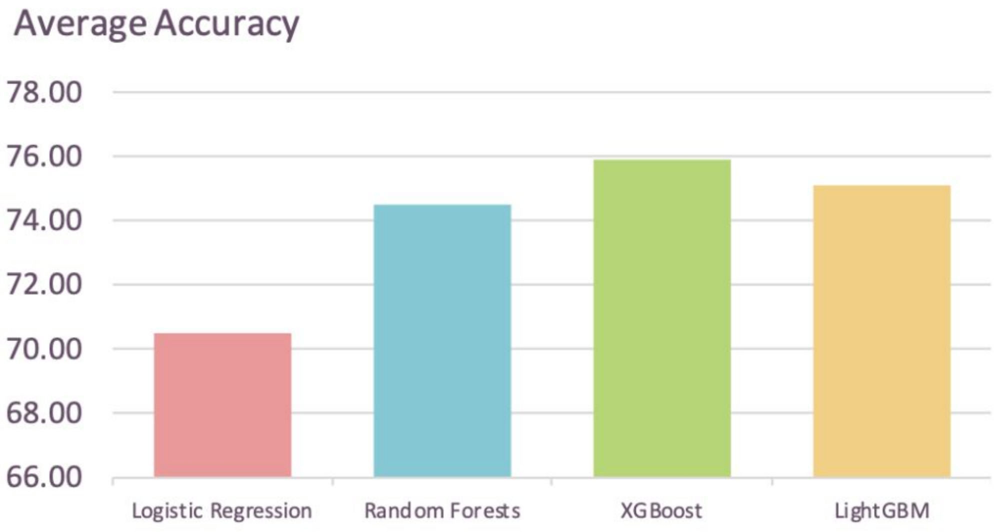


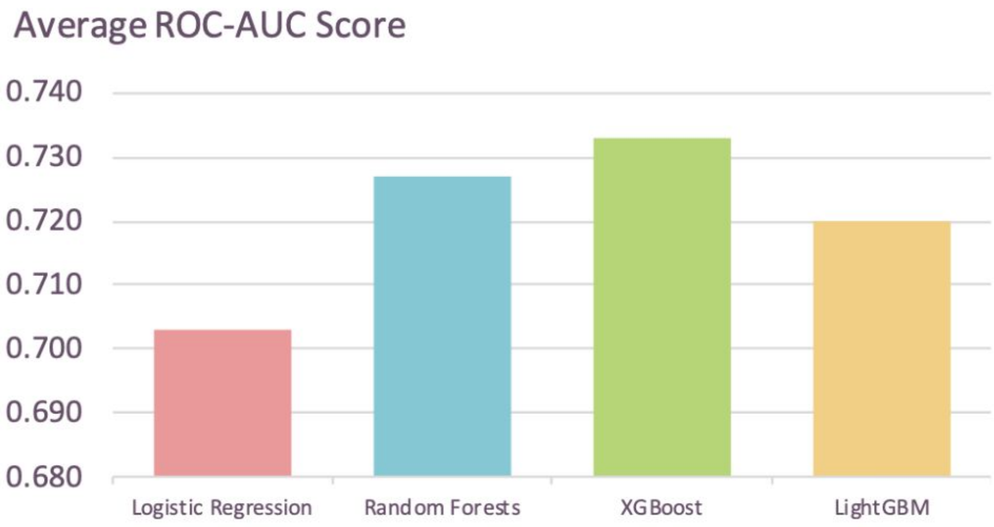
Modeling and Evaluation



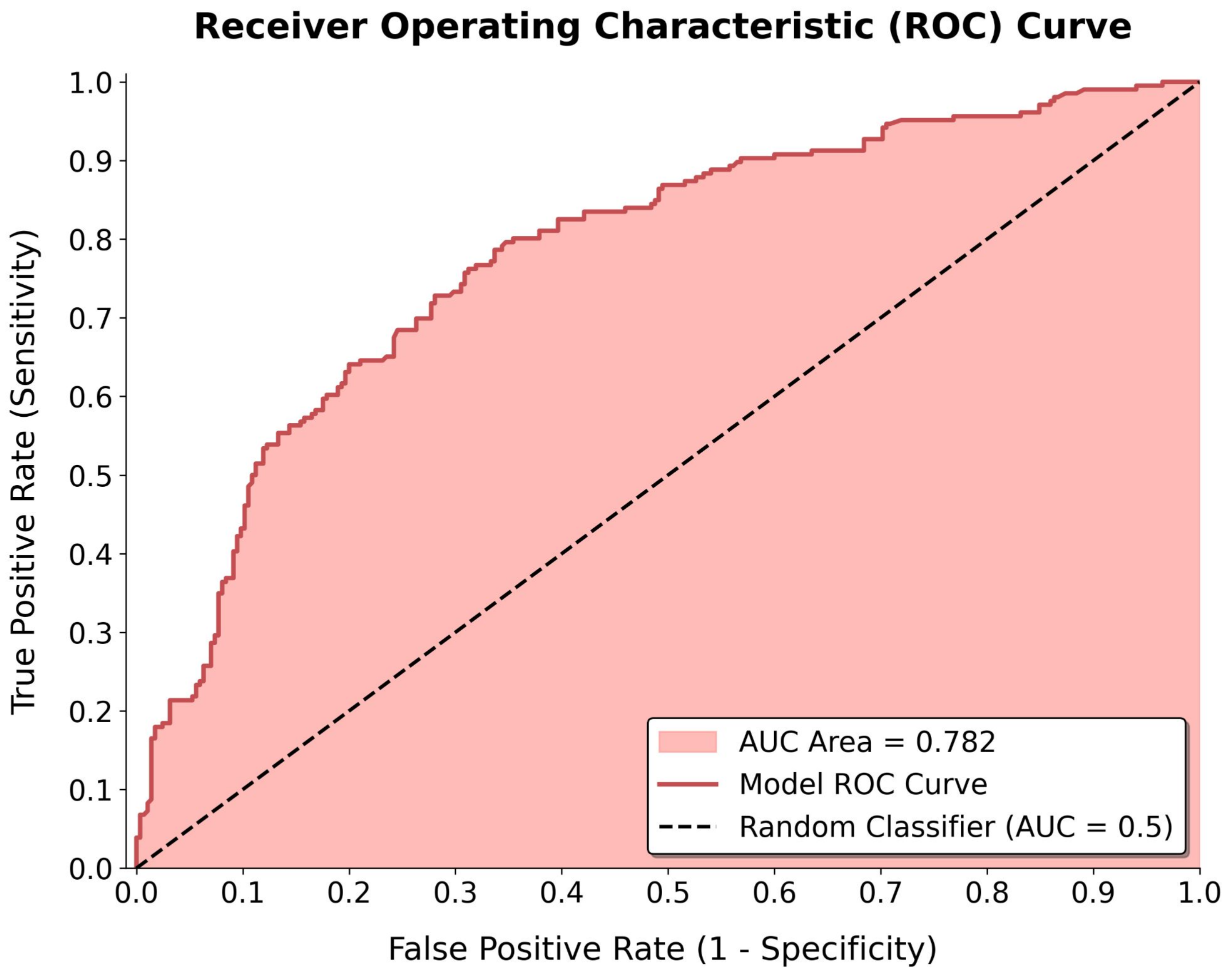
Modeling and Evaluation

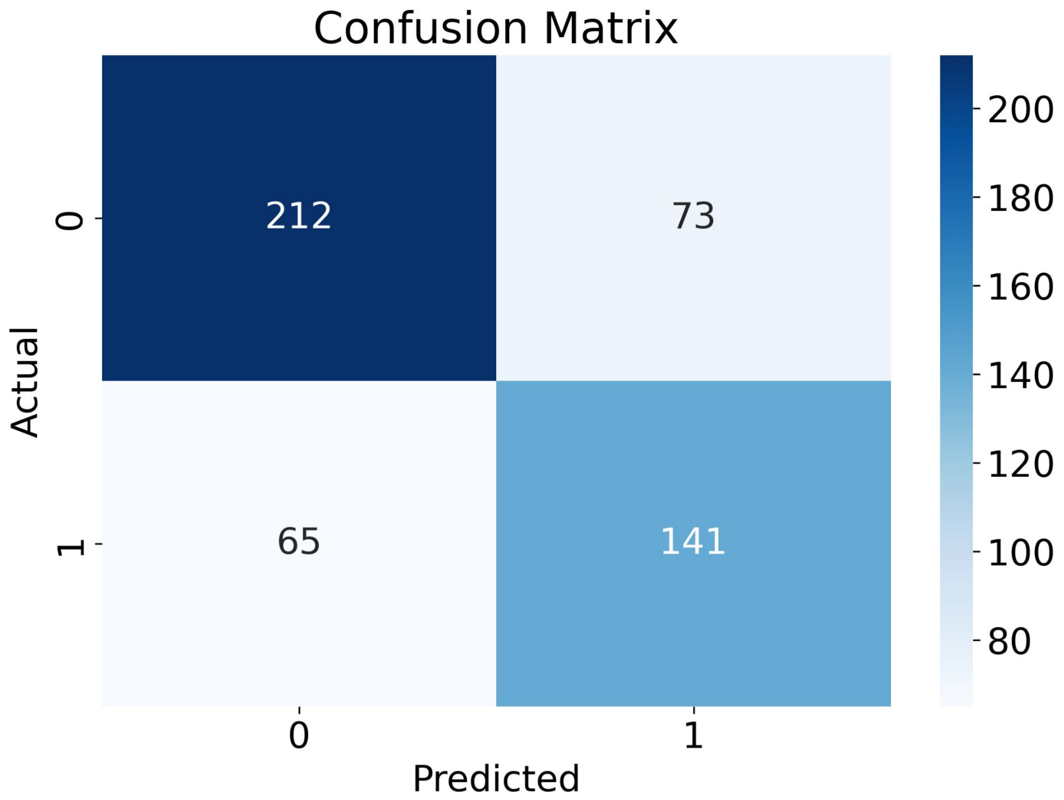
Model	Average Accuracy	Average F1-Score	Average ROC-AUC Score
Logistic Regression	70.50	0.633	0.703
Random Forests	74.50	0.658	0.727
XGBoost	75.90	0.658	0.733
LightGBM	75.10	0.646	0.720



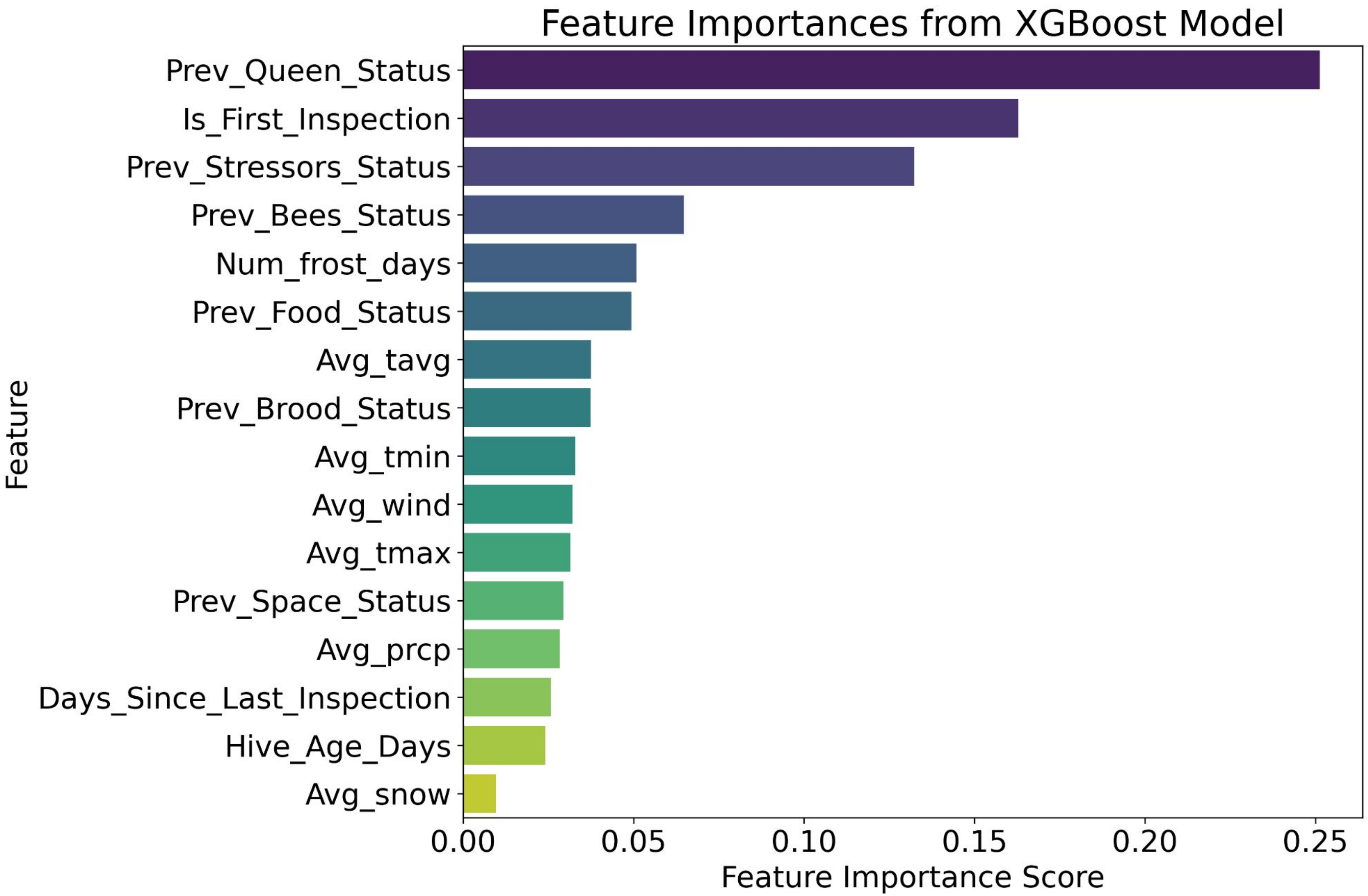


Results and Conclusion





Results and Conclusion



Limitations

- **Geographic & Climatic Limitation:** The model was trained only on data from North Carolina and Utah, so its performance in other key regions (e.g., Florida, California) is untested and likely poor.
- **Temporal & Anomaly Limitation:** The data represents only a narrow 4-year snapshot (2016-2019) and does not include major anomalous climate events (like widespread drought), limiting its ability to predict under extreme conditions.

Future Directions

- **Deeper Feature Engineering:** Experiment with different weather lag periods (e.g., 3-day acute stress vs. 21-day forage impact) to find new signals.
- **Integrate Pathogen Data:** Add hive-level **Varroa mite counts** and virus swabs. We predict this will be an extremely powerful feature, likely surpassing weather.
- **Automate with Sensors:** Evolve the model from a diagnostic to a real-time tool by replacing manual inspection features with data from in-hive sensors (acoustics, humidity, CO_2).
- **Scale "Bottom-Up":** Deploy the hive-level model across more states and aggregate the individual predictions. This will create the granular, predictive national-level tool that our "top-down" Phase 1 model could not.